

Data Bias in Crowdsourced Datasets

While "more data" is often the mantra, in computer vision—especially for agricultural applications where "green-on-green" (weeds among crops) is the primary challenge—**quality and representativeness** trump quantity every time.

To ensure your dataset is robust, we must address the "hidden" biases that often plague volunteer-collected data. Below are the key categories of bias and the literature/datasets that illustrate how to navigate them.

Crowdsourcing Bias Risk examples (Botella et al., 2018): *“Charismatic species appear to attract more visitors and foster pro-conservation behaviors [62]. Among the iNaturalist observations, Insecta, Plantae, and Aves were the most frequently recorded groups. Birdwatching, which accounted for the highest number of animal-related records, is especially popular in urban areas due to its psychological benefits and positive effects on human well-being [63]. However, iNaturalist user preferences often exhibit biases toward visually striking or culturally significant species. This leads to the over-representation of charismatic fauna and flora, such as large mammals and colorful birds, while less conspicuous organisms, like fungi and invertebrates, are under-represented [64,65]. Geographical factors, such as road density and accessibility, also influence observation patterns [66].”*

How to help volunteers: Protocol, Golden Set and Cheat Sheets

Create a **"Golden Set"**—a 1-page visual cheat sheet showing examples of "Perfect," "Acceptable," and "Rejected" photos. Volunteers respond much better to visual "Yes/No" examples than to dense text.

1. Framing & Composition (Avoiding Spatial Bias)

The AI needs to see the weed's architecture and its relationship to the surrounding environment.

- **The "Nadir" (Top-Down) Shot:** Capture at least one photo looking directly down at the weed. This mimics the perspective of a robotic weeder or drone.
- **The 45-Degree Shot:** Capture one photo from a side angle to show the plant's height and stem structure.
- **Don't "Choke" the Subject:** Leave a "buffer zone" of soil or crop around the weed (roughly **20% of the frame**). If the weed is cropped too tightly, the model won't learn how to distinguish the weed's edges from the background.
- **Center the Target:** Ensure the primary weed is in the center of the frame, even if other plants are visible.

2. Lighting & Color (Normalizing Sensors)

Different phone cameras interpret color differently. We need a "ground truth" for color and scale.

- **The Reference Object:** Place a **standardized object** (e.g., a specific coin or a provided color-calibration card) on the ground next to the weed—but not touching it.
 - *Why?* This allows us to programmatically adjust for lighting shifts and calculate the actual size (biomass) of the weed.
- **Avoid Harsh Shadows:** If possible, photograph on **overcast days** or during the "Golden Hours" (early morning/late afternoon). If the sun is high, use your body to cast a consistent shadow over the entire plant to avoid high-contrast "hot spots."
- **No Flash:** Integrated flashes flatten the image and create artificial glints on waxy leaves, which confuses texture recognition.

3. Background & Context (Eliminating "Noise")

We want the AI to learn *plants*, not *human artifacts*.

- **Clean the Frame:** Remove any non-natural debris (trash, tools, or your own feet/shadow) from the shot.
- **Agricultural Context:** Take photos of weeds *among* crops, not just on bare paths. We need the model to solve the "green-on-green" challenge.
- **Natural Variety:** Do not pull the weed or "pose" it. Capture it exactly as it is growing.

4. Growth Stages & Diversity (Reducing Frequency Bias)

A weed looks very different as a seedling than it does when flowering.

- **The Life Cycle Challenge:** We specifically need photos of **seedlings** (2–4 leaf stage). This is when they are hardest to identify but most important to kill.
- **Multiple Specimens:** Don't just photograph the biggest weed in the field. Find the small ones, the wilted ones, and the ones partially covered by crop leaves.

5. Technical Specifications

- **No Digital Zoom:** Tell volunteers to move the phone closer rather than using pinch-to-zoom. Digital zoom destroys the pixel-level texture the AI uses for classification.
- **Enable Metadata:** Ensure "Location Services" (GPS) and "Time/Date" are enabled in the camera settings. This helps us account for regional variations and seasonal growth patterns.
- **Focus Lock:** Tap the screen on the center of the weed to ensure the leaf texture is sharp, not the soil in the background.

Quick "Dos and Don'ts" for Volunteers

DO	DON'T

Include a coin for scale.	Don't hold the weed in your hand.
Take photos in cloudy weather.	Don't use digital zoom.
Get shots of tiny seedlings.	Don't crop the photo too tightly.
Keep the phone level with the ground.	Don't include your shoes or shadow.

Biases in Crowdsourced image datasets

1. Distribution & Frequency Bias

Volunteer data often suffers from **sampling bias**, where certain species are over-represented because they are "prettier," easier to find, or located in accessible areas (e.g., near roads or in gardens rather than deep in crop rows) (Hu et al., 2020; Kwon et al., 2025).

- **Relevant Literature:**
 - **Botella et al. (2018):** This study evaluated the **PI@ntNet** dataset and found a massive bias toward cultivated specimens. For agritech, a weed photographed in a backyard has a completely different background (soil, mulch, pavers) than a weed in a cornfield, leading to models that fail in actual agricultural settings (Botella et al., 2018).
 - **Engstrom et al. (2020):** They demonstrate how even "standard" datasets like ImageNet suffer from statistical bias during replication, where "selection frequency" (how often humans pick an image as a good example) can lead to models that only recognize "obvious" or "easy" versions of a class (Engstrom et al., 2020).
- **Best Practice:** Use **stratified sampling** for your collection. Instruct volunteers to seek out specific growth stages (e.g., early-stage seedlings), as identifying weeds early is more valuable for precision spraying (Saleem et al., 2022).

2. Capture & Resolution Bias

This is the "sensor gap." Volunteers will use everything from a 2018 iPhone to the latest high-end DSLR. If all your "Target Weed A" photos are high-res and "Target Weed B" are blurry, the model might learn to classify based on **image sharpness** rather than plant morphology.

- **Example Dataset: DeepWeeds**
 - The **DeepWeeds** dataset is a gold standard here. It contains 17,509 images across eight weed species in situ (Olsen, 2020). It illustrates the complexity of real-world backgrounds (dry grass, mud, varying light) and the importance of using standardized tools like **WeedLogger** to collect consistent metadata alongside photos (Olsen, 2020; Saleem et al., 2022).
- **Best Practice:** Require metadata for every upload (GPS, time of day, camera model). Encourage volunteers to use a **reference object** (like a coin or a standard color card) in some shots to help the model learn scale and normalize color across different sensors (Sharma et al., 2024).

3. Semantic & Labeling Bias

In crowdsourcing, non-experts often confuse similar-looking species or use different names for the same plant (e.g., "forest" vs. "trees"). This is known as the **sensory gap** (Murillo Montes de Oca et al., 2017).

- **Relevant Literature:**
 - **Yasmin et al. (2022):** This work suggests using "expanded input elicitation." Instead of just asking "Is this a weed?", ask volunteers for their **confidence level** or to click the specific location of the weed in the frame. Aggregating these multi-modal inputs significantly improves ground-truth reliability (Yasmin et al., 2022).
 - **Hu et al. (2020):** Proposed a human-in-the-loop workflow where crowd workers are used not just to label, but to **detect bias** in the dataset by finding similarities that shouldn't be there (e.g., "all images of this weed have a shadow") (Hu et al., 2020).

Summary of Open Datasets for Benchmarking

Dataset	Focus	Scale / Diversity	Key Lesson
iNaturalist	Global Biodiversity	Millions of images	Shows geographic and seasonal bias (Kwon et al., 2025).

DeepWeeds	Australian Rangelands	17.5k images, 8 classes	Excellent for in-situ complexity (Olsen, 2020).
PlantVillage	Crop Diseases	50k+ images	Warning: Mostly "lab-style" (clean backgrounds), poor generalization (Botella et al., 2018).
SugarCane/Weed	Indian Agriculture	High-res UAV + Ground	Illustrates "camouflage" bias (weeds looking like crops) (Singh et al., 2024).

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